Probabilistic Feature Imputation and Uncertainty-Aware Multimodal Federated Aggregation

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Code & Models

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The Problem

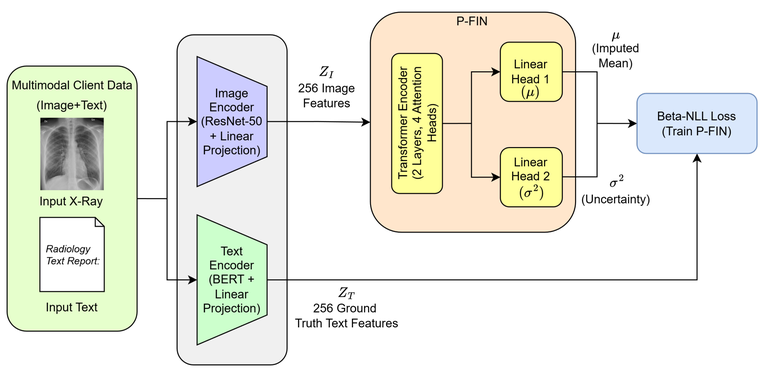
- Image + text models beat image-only diagnosis, but privacy law forces **federated** training — data never leaves the hospital.
- **Modality heterogeneity:** large centers hold image + report; small clinics hold *image only*.
- Existing imputers are **deterministic** — one confident guess that can hallucinate a report, and every imputed feature is trusted equally.

Contributions

Impute the missing modality as a **distribution**, not a point — then let the predicted uncertainty drive both local fusion and global aggregation.

- **P-FIN** — calibrated per-dimension uncertainty via β-NLL.
- **Local gating** — suppress unreliable dimensions before fusion.
- **Fed-UQ-Avg** — weight clients by confidence, not size alone.
- **+5.36% AUC** over deterministic FIN+FedAvg; calibration error ECE = 0.0422.

Stage 1 — P-FIN Training

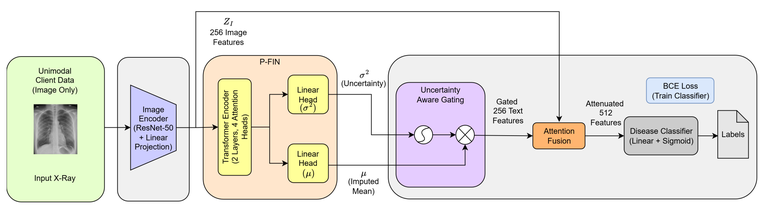


P-FIN maps image features to a text-embedding *distribution* (μ, σ^2), trained with β-NLL for calibrated uncertainty.

ResNet-50 and BERT encoders give L2-normalized 256-d features. Dual heads emit $\mu, \sigma^2 \in \mathbb{R}^{256}$; stop-gradient β-NLL ($\beta = 0.5$) prevents variance explosion:

$$\mathcal{L} = \frac{1}{d}\sum_j \mathrm{SG}(\sigma_j^{2\beta})\left(\frac{1}{2}\log\sigma_j^2 + \frac{(z_j^T-\mu_j)^2}{2\sigma_j^2}\right).$$

Stage 2 — Uncertainty-Aware Fusion



The uncertainty gate attenuates unreliable dimensions of μ before cross-modal attention fusion and classification.

Gate $g = \mathrm{sigmoid}(-\log\sigma^2)$: high uncertainty $\Rightarrow g \to 0$, damping noisy dimensions *before* fusion, so a classifier never overfits to hallucinated features.

Global Aggregation — Fed-UQ-Avg

FedAvg weights clients by data size alone, letting a large but poorly-imputing site dominate. Fed-UQ-Avg blends size with imputation **confidence** ($\alpha = 0.6,\ T = 0.2$):

$$\mathrm{conf}_k = \exp\left(-\frac{\bar{\sigma}_k^2}{T}\right)$$

$$W_k = (1-\alpha)\, w_k^{\mathrm{data}} + \alpha\, w_k^{\mathrm{conf}}$$

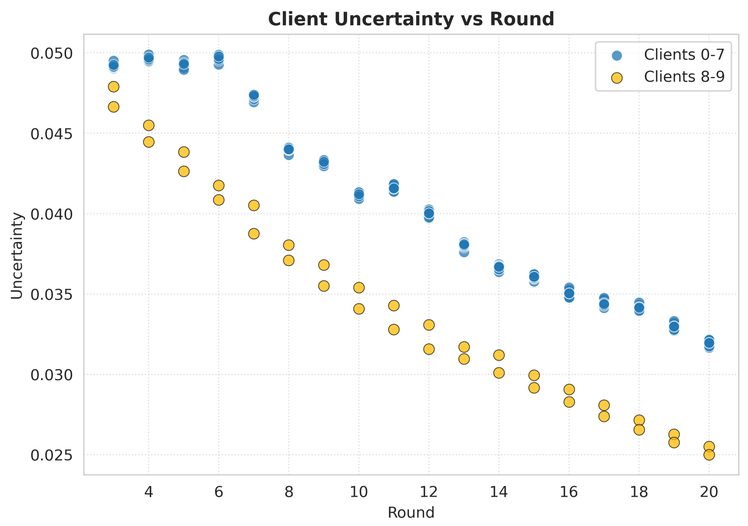
Confident clients are up-weighted; unreliable ones contribute less to the global model.

Setup

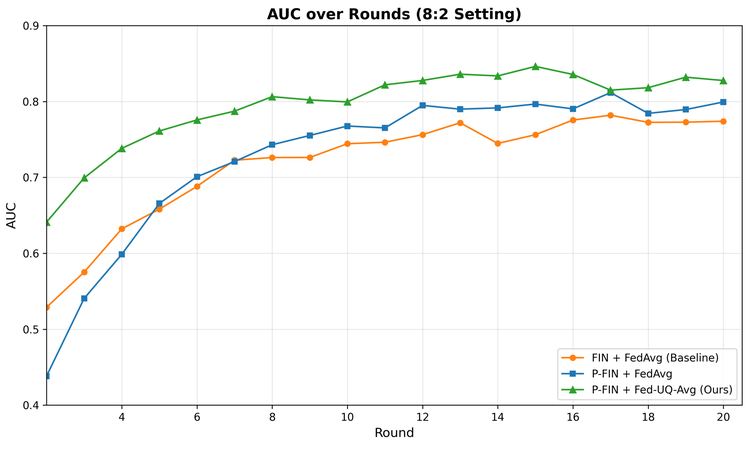
- **Data:** CheXpert (unimodal), NIH Open-I (multimodal); PadChest held out for external validation.
- **Federation:** $K = 10$ clients; unimodal : multimodal $= 8{:}2,\ 6{:}4,\ 4{:}6$; non-IID Dirichlet ($\alpha_{\mathrm{Dir}} = 0.5$).
- **Training:** 20 rounds $\times$ 4 local epochs; Adam, lr 10^{-4}, batch 32; backbones projected to $d = 256$.

Convergence

A calibrated uncertainty signal is only useful if it *sharpens* as the model learns — and only worthwhile if that sharpening shows up in accuracy. The two curves below establish both, in the hardest 8:2 regime.



Each unimodal client's mean imputation uncertainty (blue) decays steadily as P-FIN absorbs the cross-modal mapping from the multimodal minority (orange); the high early variance is precisely what discounts unreliable first-round imputations during aggregation.



Both P-FIN variants pull away from the FIN+FedAvg baseline; the gap between the top two curves is the Fed-UQ-Avg aggregation gain.

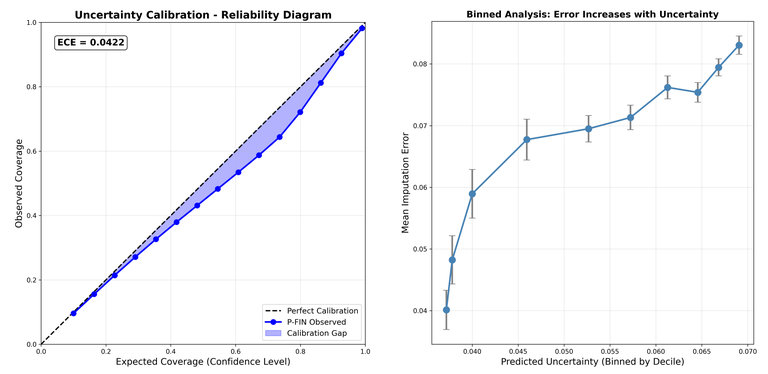
Results — Test AUC (%)

Method	8:2	6:4	4:6
Zero-filling	72.56	75.61	77.86
Uniform-filling	71.78	74.18	76.67
FIN + FedAvg	77.40	79.31	81.28
P-FIN + FedAvg	79.93	81.72	83.11
Ours (P-FIN + Fed-UQ-Avg)	**82.76**	**84.14**	**85.89**

Ours leads at every ratio. Ablation: Fed-UQ-Avg adds **+2.83 / +2.42 / +2.78%** over P-FIN+FedAvg — largest where text is scarcest.

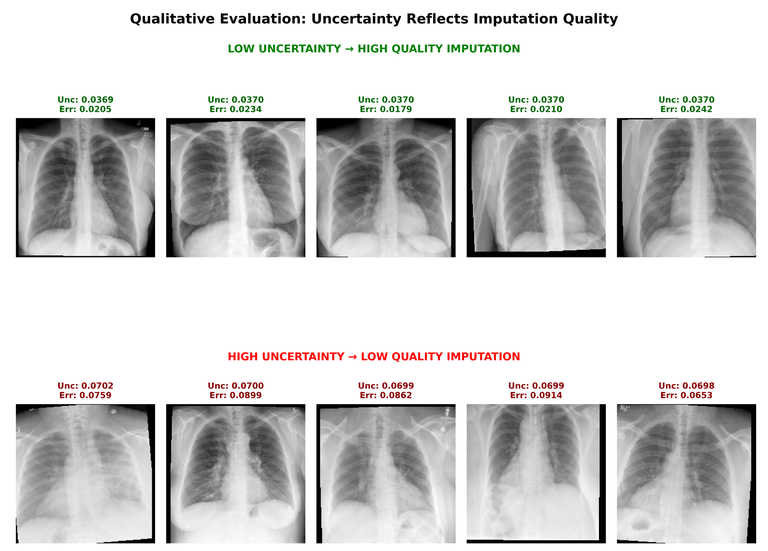
+5.36%
AUC over deterministic FIN+FedAvg in the 8:2 regime
(80% of clients have no text)

Calibration



Reliability diagram tracks the diagonal (ECE = 0.0422, left); mean imputation error rises **monotonically** across uncertainty deciles (right).

Qualitative



Low-uncertainty cases correspond to clear imaging; high-uncertainty cases correspond to complex or ambiguous presentations.

Takeaways

- Calibrated **distributions** instead of point estimates yield a "learning to distrust" paradigm for federated medical imaging.
- Gains are **largest where data is scarcest** — the regime that matters for under-resourced clinics.
- Aligns with **FDA** priorities for uncertainty reporting in AI-enabled medical devices.
- *Next:* bidirectional synthesis and epistemic uncertainty for out-of-distribution robustness.